INCH-POUND

MIL-F-20420J(AR)
AMENDMENT 4
12 March 1993
SUPERSEDING
AMENDMENT 3
20 May 1991

MILITARY SPECIFICATION

FLARE, SURFACE, TRIP M49A1, PARTS,
AND LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-F-20420J(AR), dated 25 March 1983, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1.1 Delete: "FEDERAL QQ-P-416 - Plating, Cadmium (Electro deposited)".

PAGE 3

2.2 Add new paragraph as follows:

"2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM - B117 "STANDARD METHOD OF SALT SPRAY (FOG) TESTING"

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)"

3.5.2 Cold temperature: Add the phrase, "except that the pull pin force requirements do not apply." to the last sentence.

3.5.3 Delete: "Temperature and Humidity Cycling" and substitute "Temperature and Humidity Cycling (First Article Only)".

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PAGE 5

Table I: Change the number of sample units on each of the items on the first page of Table I from "25" to "10".

PAGE 6

Table I: Change the number of sample units on each item on the second page of Table I from "25" to "10".

PAGE 7

Table I: Paragraph Reference for the Flare Assembly - Examination of Defects: Change from "4.4.2.19" to "4.4.2.18, 4.4.2.19, 4.4.2.20, 4.4.2.21 and 4.4.2.22 (except major defect 105 thru 109)".

Change the number of sample units for the examination of defects for the Trigger Spring, Retainer Spool, and the Wire Spool Assembly from "25" to "10".

PAGE 9

4.4.2.1 Delete major defect 102 and substitute: "102 Thickness of bottom excluding angular surface, min. 0.40% 3.2 Gage".

Delete major defect 105 and minor defect 202 in their entirety.

PAGE 10

4.4.2.2 Delete major defect 109 and substitute: "109 Angle of turn-up at end of handle, min. 0.40% 3.2 Gage".

Add major defect 110: "110 Protective coating test 4.4.3.7, 4.4.3.7, 3.2, 4.4.3.7".

Delete minor defects 204, 205, and 206 in their entirety.

PAGE 11

4.4.2.2 Add minor defects:

"214 Presence of dimple (2 places) 0.65% 3.2 Visual".

"215 Height of turn-up at end of handle, min. 0.65% 3.2 Gage".

Delete minor defects 208, 209, 210, and 211 in their entirety.

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PAGE 12

4.4.2.3 Delete major defect 103 and substitute: "103 Location of firing pin hole 0.40% 3.2 Gage".

Delete minor defect 203 in its entirety.

PAGE 13

4.4.2.4 Delete major defect 101 and substitute: "101 Location of point 0.40% 3.2 Gage".

Delete major defect 103 and substitute: "103 Point profile damaged 0.40% 3.2 Gage".

Add major defect 105: "105 Protective coating test 4.4.3.7 4.4.3.7 3.2 4.4.3.7".

PAGE 15

4.4.2.6 Delete minor defect 201 and substitute: "201 Length from top edge of bracket to top edge of slot 0.65% 3.2 Gage".

Delete minor defect 203 and substitute: "203 True position of carriage bolt holes (round and square) 0.65% 3.2 Gage".

Delete minor defect 204 and substitute: "204 True position of trigger pivot hole 0.65% 3.2 Gage".

Add Note 1 callout to minor defect 205.

Delete minor defect 207 and substitute: "207 Width of mounting surface, min 0.65% 3.2 Gage".

Add the following under notes:

"1 Surface imperfections which cause the local material thickness to be below minimum shall be classed defective. Cracks appearing on the top lugs or the tabs of the "E" slot will not be permitted."

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PAGE 16

4.4.2.7 Add major defect 106: "106 Protective coating test 4.4.3.7 4.4.3.7 3.2 4.4.3.7".

Add Note 1 callout to major defect 104.

Add the following under notes:

1 Surface imperfections in the finger area which result in a local thickness or width that is less than the minimum required shall be classed defective."

PAGE 18

4.4.2.9 Add major defect 102:

"102 Protective coating test 4.4.3.7 4.4.3.7 3.2 4.4.3.7".

PAGE 19

4.4.2.10 Delete major defect 101 and substitute: "101 Diameter of wire (before forming) 0.40% 3.2 Gage".

PAGE 20, 21 and 23

Delete in their entirety and substitute new page 20, 21 and 23 included.

PAGE 22

4.4.2.13 Delete major defect 101 and substitute: "101 Diameter of press fit surface 0.40% 3.2 Gage".

Delete major defect 102 and substitute: "102 True position of hinge pin hole(s) 0.40% 3.2 Gage".

Delete major defect 105 and substitute: "105 Length from primer hole to lever forward pivot 0.40% 3.2 Gage".

Delete major defect 113 and substitute: "113 Length of press fit surface to taper 0.40% 3.2 Gage".

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Spring, Striker	1 OF	8864755
			NEXT HIGHER ASSEMBLY: 8836957, 9288866 or 9313553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Embrittlement of spring	10	0-1	3.3	4.5.1
102	Torque test	50	0-1	3.2	4.5.8
103	Protective coating test	4 4.3.7	4.4.3.7	3.2	4.4.3.7
Minor					
201	Outside diameter	1/	0.65%	3.2	Gage
202	Length over long leg	1/	0.65%	3.2	Gage
203	Length of long leg hook		0.65%	3.2	Gage
204	Diameter of wire (before forming and plating)		0.65%	3.2	Gage
205	Length over short leg	1/	0.65%	3.2	Gage
206	Length from long leg to short leg		0.65%	3.2	Gage
207	Length over bend of short leg	1/	0.65%	3.2	Gage
208	Length of short leg hook		0.65%	3.2	Gage
209	Overall length	1/	0.65%	3.2	Gage
210	Poor workmanship		1.00%	3.8	Visual

NOTES:
1/ See paragraph 4.5.13 for alternate method of dimensional control.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.12	Safety Clip Assembly		8836958
			NEXT HIGHER ASSEMBLY: 8836957, 9288866 or 9313553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of wire (prior to forming and plating)		0.40%	3.2	Gage
102	Crack		0.40%	3.2	Visual
103	(Deleted)				
104	Irregular plating of pull pin		0.40%	3.2	Visual
105	Safety clip assembly spring test	4.4.3.5	4.4.3.5	3.2	4.5.7
106	Protective coating test	4.4.3.7	4.4.3.7	3.2	4.4.3.7
Minor					
201	Length from bend to end, min.		0.65%	3.2	Gage
202	Length		0.65%	3.2	Gage
203	Pull pin incorrectly assembled		0.65%	3.2	Visual
204	Width at closed end improper		0.65%	3.2	Gage
205	Spring clip safety missing or mislocated		0.65%	3.2	Visual
206	Poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.14	Spring, Trigger	1	1	8836975

NEXT HIGHER ASSEMBLY: 8836973

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Embrittlement of spring	10	0-1	3.3	4.5.1
102	Spring wound improperly		0.40%	3.2	Visual
103	Protective coating test 4.4.3.7	4.4.3.7	4.4.3.7	3.2	4.4.3.7
Minor					
201	Inside diameter of spring	1/	0.65%	3.2	Gage
202	Inside diameter of loop, max.		0.65%	3.2	Gage
203	Distance from loop centerline to tangent of free end (free state)		0.65%	3.2	Gage
204	Width of spring center to loop center		0.65%	3.2	Gage
205	Length of free leg to spring center	1/	0.65%	3.2	Gage
206	Width of free end	1/	0.65%	3.2	Gage
207	Entire loop not flat		0.65%	3.2	Gage
208	Diameter of wire		0.65%	3.2	Gage
209	Poor workmanship		0.65%	3.8	Visual

NOTES:
1/ See paragraph 4.5.13 for alternate method of dimensional control.

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Add major defect 114: "114 Protective coating test 4.4.3.7 4.4.3.7 3.2 4.4.3.7".

PAGE 24

4.4.2.15: Delete major defect 106 and substitute: "106 Spring test 4.4.3.9 3.2 4.5.12".

Add major defect 107: "107 Protective coating test 4.4.3.7 4.4.3.7 3.2 4.4.3.7".

PAGE 27

4.4.2.18: Illuminant Assembly. Delete Major 101 and substitute: "Interior of case not coated with insulating composition to within the specified limit".

PAGE 31

4.4.2.22: Delete major defect 105 and substitute: "105 Jumble test 4.4.3.1 3.4 4.5.2".

Delete "108 Functioning (Temperature and Humidity Cycling) 4.4.3.3.3 3.5.3 4.5.4.3."

PAGE 37

* 4.4.2.27: Box, Packing. Minor 204, delete "ICC".

PAGE 39A

4.4.2.30 Add new page 39A included.

PAGE 40

Delete the following from Table III Asbestos: "(9269018 and 9269105)".

PAGE 41

4.4.3.2 Delete in the last sentence: "4.4.3 and substitute "4.5.3".

4.4.3.3: Delete: "Functioning. The functioning test consists of three(3) distinct phases: Ambient, Cold, and Temperature-Humidity Cycling. Reduced....in 4.5.4" and substitute "Functioning. The functioning test consists of two phases Ambient and cold. Reduced....in 4.5.4."

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4.4.3.3.3: Delete in its entirety.

4.4.3.3.3.1: Delete in its entirety.

4.4.3.3.3.2: Delete in its entirety.

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CLASSIFICATION OF DEFECTS & TESTS

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AMENDMENT 4

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.30	Trigger Washer		8836977
			NEXT HIGHER ASSEMBLY 8836973

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Protective coating test	4.4.3.7	4.4.3.7	3.2	4.4.3.7
102	Inside diameter, min		0.40%	3.2	Gage
Minor					
201	Poor workmanship		1.00%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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PAGE 45

4.4.3.7: Delete paragraph in its entirety and substitute the following:

"4.4.3.7 Protective coating (salt-spray) test of clip, safety assembly (dwg. 8836958), lever (dwg. 8836961), cover (dwg. 8836965), striker assembly (dwg. 8836971), pivot, trigger (dwg. 8836974), spring, trigger (dwg. 8836975), trigger (dwg. 8836976), washer, trigger (dwg. 8836977), spring striker (dwg. 8864755), and spool retainer - major defects. Testing shall be performed after application of all specified protective finishes. The Salt Spray Test shall be conducted in accordance with ASTM B117, "Standard Method of Salt Spray (Fog) Testing," for a period of 96 hours. To secure uniformity of test results, all test samples shall be aged at room temperature for 24 hours before being subjected to the Salt Spray. A quantity of four parts from each production lot shall be used as the test samples in lieu of specimens. The salt spray tested samples shall show no evidence of white corrosion products (if cadium or zinc is present) or base metal corrosion products at the end of the 96 hour test. The appearance of corrosion products (visible to the unaided eye at normal reading distance) on one or more the test samples shall be cause for rejection, except that corrosion products at the edge of an unprotected area shall not be cause for rejection."

PAGE 46

Change title of paragraph 4.5.3.1 from "Baffle and asbestos: to: "Baffle".

PAGE 47

4.5.4.2: Add new last sentence: "Pull pin force shall be recorded for information only."

4.5.4.3: Delete "Temperature and Humidity Cycling. The flare...(Destructive test)" and substitute: "Temperature and Humidity Cycling (First Article Only). The flare... (Destructive test). Pin pull force shall be recorded for information only."

PAGE 48

Change title of paragraph 4.5.9.1 from; "Intermediate change and first composition" to "Intermediate charge and first fire composition".

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PAGE 50

Delete paragraph 4.5.10 in its entirety.

Add the following paragraph:

"4.5.13 Dimensional control of formed parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the forming process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each day's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer."

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